

ActionShap: Evaluating Recommendation Attributions under Bounded Profile Interventions

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Abstract

Recommendation explanations are commonly evaluated by deleting historical interactions, although deployed recommendation systems act through bounded changes to the influence of observed evidence. Deletion and a declared system-executable intervention therefore measure different properties. We introduce **ActionShap**, a model-interface-agnostic protocol for history-conditioned recommenders exposing inference-time profile weighting. ActionShap separates deletion alignment, bounded-intervention alignment, and the quality of the joint action selected under a fixed budget. It fixes temporal splits, complete pre-test-history exclusion, target-plus-unseen-negative candidate sets, global tie priorities, signed action selection, abstention, convergence, and inference over distinct users. We evaluate Monte Carlo Shapley, locally weighted LIME, leave-one-out, greedy sequential deletion, and a random control on MovieLens-1M and Amazon Digital Music. The primary analysis uses a history-conditioned ItemKNN recommender, 1,000 users, five common seeds, and exact budget-two oracles; a latent profile model and full-unseen-catalogue subset provide robustness evidence. Shapley exhibits a positive bounded-minus-deletion alignment difference in the primary ItemKNN conditions, especially on Amazon. LIME and leave-one-out retain higher absolute bounded alignment, and on MovieLens they achieve slightly higher realized NDCG effects and lower conditional normalized regret than Shapley. The random control remains near its within-user null. ActionShap contributes an auditable evaluation instrument for separating perturbation sensitivity, absolute alignment, and intervention decision quality rather than claiming universal superiority for any explainer.

Keywords: Explainable recommendation, Intervention evaluation, Faithfulness, Actionability, Shapley value, LIME

1 Introduction

An explanation is often consumed as advice about where to act. A user, analyst, or system operator does not usually ask only whether a recommendation score changes when an input is deleted; they ask what will happen if a factor is discounted, suppressed, or otherwise modified within an available operating policy. The distinction is important in recommender systems because historical interactions are evidence, not necessarily controls. Removing an interaction can be a useful faithfulness diagnostic while being an unrealistic intervention.

This paper studies the resulting evaluation gap. We ask whether an attribution ranking predicts the effect of a feasible modification to the interaction history, and whether the explanation leads to a useful joint intervention under a fixed budget. The question is deliberately narrower than recourse: ActionShap does not claim that a user can cause a profile change, does not estimate causal effects in the world, and does not propose a new recommender architecture. It audits an explanation inside a frozen recommender and an explicitly declared intervention simulator.

The protocol is motivated by three observations. First, deletion and bounded modification are different estimands. A deletion moves a factor to a baseline that may be far outside an operational range. Second, pointwise attribution quality does not determine decision quality: a method can rank individual effects well while selecting a harmful pair, or select a good pair despite modest average rank agreement. Third, a method comparison is not interpretable without a chance control, valid-user counts, convergence diagnostics, and inference over distinct users rather than repeated seed-user records.

ActionShap addresses these issues with a history-conditioned interaction-player game. For each user, the most recent training interactions form the player set. A frozen ItemKNN model computes scores from the retained history at inference time, allowing masking and bounded downweighting to change the profile without retraining. The primary characteristic function is a continuous target-margin utility selected by an independent convergence study; NDCG@10 remains the operational ranking outcome. Every method receives the same temporal users, candidates, intervention strengths, action space, and tie rules.

Our contributions are fourfold:

1. We formalize a recommendation-specific evaluation protocol that separates deletion faithfulness, bounded intervention, and budgeted decision quality under a frozen model.
2. We provide complementary magnitude, signed-direction, success, abstention, regret, stability, and within-user null metrics, with explicit handling of constant vectors and inactive oracles; harmful actions are represented by their negative realized effect rather than a separate headline aggregate.
3. We give executable algorithms for paired Monte Carlo attribution, signed action selection, the exact $B \leq 2$ oracle, user-level inference, and independent convergence selection.

4. We report a two-dataset, two-model, five-seed benchmark. The primary ItemKNN results show that Shapley can improve relative to deletion while LIME and LOO retain higher absolute alignment and downstream decision quality in some conditions.

The paper makes no a priori claim that Shapley, LIME, LOO, or any other explainer is best. The framework is the contribution; method rankings are empirical and bounded by the declared offline simulator.

2 Literature Review

2.1 Explainable recommendation

Explainable recommendation includes feature, item, path, attention, sentiment, and natural-language rationales. Surveys emphasize that these forms serve different objectives, including user persuasion, model fidelity, transparency, and preference communication [1]. Latent-factor explanations expose item or feature contributions, while post-hoc methods apply model-agnostic surrogates such as LIME and SHAP [2–5]. Sentiment- and phrase-level models construct explanations during recommendation rather than evaluating a post-hoc attribution [6]. ActionShap addresses a complementary question: whether an explanation predicts the outcome of a declared feasible change.

2.2 Recommender models and offline evaluation

Classical matrix factorization and pairwise learning-to-rank remain foundations for offline recommendation [7–10]. Neural collaborative filtering, recurrent session models, self-attentive sequence models, and bidirectional transformer recommenders broaden the architecture space [11–14]. Graph recommenders introduce a further distinction between a trained graph representation and an inference-time profile intervention [15, 16]. We use ItemKNN as the primary model because its history dependence is transparent under masking; the profile aggregator is a robustness model, not a claim to cover all architectures.

Offline evaluation depends on temporal leakage, negative sampling, exposure, and the candidate universe. Results on a target-plus-negative set are sampled-ranking results, not catalogue retrieval. Evaluation choices can materially change rankings between methods and models [17–19]. ActionShap fixes these choices before final testing, reports popularity references, and separates sampled ranking from full-unseen-catalogue evaluation.

2.3 Faithfulness, counterfactual evaluation, and recourse

Deletion and insertion tests measure model dependence but can be sensitive to the baseline and perturbation path. Sanity checks show that some attribution maps can appear plausible while failing model or data randomization tests [20]. Perturbation-based explanations also raise the question of whether the perturbation remains within a meaningful data manifold [21]. Counterfactual recommendation and recourse methods instead search for profile or graph changes that alter a ranking or decision [22–32].

Table 1 Positioning by evaluation target.

Approach	Explainer audit	Feasible policy	Joint regret	Abstention
Deletion/insertion	Yes	No	No	No
Counterfactual explanation	Usually generates	Method-specific	Sometimes	Method-specific
Algorithmic recourse	No	Yes	Optimization target	Often
ActionShap	Yes	Yes	Yes	Yes

ActionShap does not generate causal recourse. It evaluates an attribution under a fixed intervention policy and makes the policy, cost, authority, and offline limitations explicit.

2.4 Interpretable machine learning and cooperative attribution

Interpretability is not a single property: faithfulness, stability, simulatability, and usefulness can disagree [33–37]. SHAP connects feature attribution with cooperative-game values, while sampling makes the construction applicable when exact enumeration is infeasible [38–41]. ActionShap treats Shapley as one interchangeable explainer. Its axioms do not imply causal validity, intervention feasibility, or decision superiority; these are measured separately.

3 Background and Problem Formulation

3.1 Temporal user-specific game

For user u , let H_u^{train} contain all interactions before validation and test. The player set is the most recent training window,

$$P_u = \{p_{u1}, \dots, p_{un_u}\}, \quad n_u \leq n_{\max} = 20, \quad (1)$$

and define the older training context $B_u = H_u^{\text{train}} \setminus P_u$. Interactions are ordered by (timestamp, original_record_index); the last event is test and the preceding event is validation. Validation and test events are never players. Models are fitted using the complete training histories H_u^{train} , but the released scoring implementation excludes older interactions B_u from inference-time scoring: the profile score $f_u(i; S)$ depends only on S . Thus B_u is not silently added as fixed scoring context. Consequently, $n_{\max}$ is both the operating profile boundary and the player-game boundary; the $n_{\max} \in \{50, 100\}$ runs are sensitivity analyses of that boundary, not experiments that hold the recommender input fixed.

Every user has a fixed candidate set E_u containing the held-out target and sampled negatives. Negatives exclude the complete pre-test history, including validation. The user seed, candidate seed, tie seed, model seed, and attribution seed are independent. Full-catalogue robustness uses the target plus all unseen items. Target coverage is one by construction and is not called retrieval recall.

3.2 History-conditioned recommenders

The primary ItemKNN score is a weighted mean of frozen item–item similarities. We write the inference-time weighted profile as $\mathcal{P}_u = (P_u, w)$ and define $W_u(S) = \sum_{h \in S} w_h$:

$$f_u(i \mid S, w) = \begin{cases} \frac{\sum_{h \in S} w_h s(i, h)}{W_u(S)}, & W_u(S) > 0, \\ 0, & W_u(S) = 0. \end{cases} \quad (2)$$

A profile aggregator is used for robustness,

$$p_u(S, w) = \begin{cases} \frac{\sum_{h \in S} w_h q_h}{W_u(S)}, & W_u(S) > 0, \\ \mathbf{0}, & W_u(S) = 0, \end{cases} \quad f_u(i \mid S, w) = q_i^\top p_u(S, w). \quad (3)$$

Thus the empty coalition and any nonempty all-zero-weight coalition use the same zero-profile convention. The primary full profile uses $w_h = 1$ for every retained interaction. For readability below, $f_u^S(i)$ abbreviates $f_u(i \mid S, \mathbf{1})$ and $s_i^S = f_u^S(i)$. Its item embeddings are trained once with the leave-one-out BPR-style objective

$$\mathcal{L}_{\text{BPR}} = - \sum_{(u, i, j)} \log \sigma(p_{u, -i}^\top (q_i - q_j)) + \frac{\lambda}{2} \left(\|q_i\|_2^2 + \|q_j\|_2^2 + \sum_{h \in H_u^{\text{train}} \setminus \{i\}} \|q_h\|_2^2 \right), \quad (4)$$

where $p_{u, -i}$ is the mean of the other complete-training-history embeddings, j is sampled uniformly outside that history, $\lambda = 10^{-4}$, the embedding dimension is 64, and ten epochs with one sampled positive per user per epoch are used. Static user embeddings are excluded from attribution because post-training history masking cannot change their scores.

3.3 Utilities and interventions

The primary characteristic function is the target-margin utility

$$v_u^{\text{attr}}(S) = \sigma \left(s_y^S - \frac{1}{L} \sum_{i \in \text{TopL}_{-y}(S)} s_i^S \right), \quad (5)$$

where $s_i^S := f_u^S(i)$, y is the temporal target, $L = 10$, and $\sigma(t) = (1 + e^{-t})^{-1}$ has unit temperature. The competitor set $\text{TopL}_{-y}(S)$ is recomputed from the current coalition scores for every coalition S ; it contains the ten highest-scoring candidates other than y . Every retained positive interaction starts with inference-time weight $w_h = 1$. No temporal decay or rating-dependent weight is used. An action changes only these inference-time weights and never retrains the model. For the fixed candidate set E_u ,

let τ_i be the global tie priority and define the one-indexed target rank

$$r_u^S(y) = 1 + \sum_{i \in E_u \setminus \{y\}} \mathbf{1}[s_i^S > s_y^S] + \sum_{i \in E_u \setminus \{y\}} \mathbf{1}[s_i^S = s_y^S \wedge \tau_i < \tau_y]. \quad (6)$$

With one relevant target, $IDCG@10 = 1$ and the operational ranking outcome is

$$q_u(S) = \text{NDCG}@10(f_u^S, y) = \mathbf{1}[r_u^S(y) \leq 10] \frac{1}{\log_2(r_u^S(y) + 1)}. \quad (7)$$

Thus $\text{HitRate}@10$ is the one-target hit indicator, not catalogue retrieval recall. All effects and regrets are stored separately for target margin and NDCG. The empty coalition has target-margin utility 0.5 and deterministic catalogue ties.

For an action $A \subseteq P_u$, let $P_u^{A,\rho}$ denote the same retained player window with weights

$$w_p^{A,\rho} = \begin{cases} \rho, & p \in A, \\ 1, & p \notin A, \end{cases} \quad (8)$$

and define the signed intervention effect for either utility $z \in \{v_u^{\text{attr}}, q_u\}$ by

$$\Delta_u^z(A; \rho) = z_u(P_u^{A,\rho}) - z_u(P_u). \quad (9)$$

Deletion and bounded singleton effects are therefore

$$\Delta_{u,p}^{z,\text{del}} = \Delta_u^z(\{p\}; 0), \quad \Delta_{u,p}^{z,\text{bnd}} = \Delta_u^z(\{p\}; 0.5). \quad (10)$$

Deletion is the diagnostic $w_p \leftarrow 0$. The primary feasible intervention is bounded downweighting,

$$\text{do}(p; \rho) : w_p \leftarrow \rho w_p, \quad \rho = 0.5, \quad (11)$$

with $\rho = 0.25$ as a predeclared sensitivity. No model retraining occurs after an intervention.

The action space is

$$\mathcal{A}_{u,B} = \{A \subseteq P_u : |A| \leq B\}, \quad (12)$$

including no action. The primary scientific budget is $B = 2$. A signed attribution predicts downweighting benefit as $-\phi$; only positive predicted benefit is selected. LOO is an algebraic deletion oracle at $B = 1$, not a headline competitor.

4 Methodology

4.1 Protocol architecture

Figure 1 summarizes the evaluation pipeline used throughout the paper. The architecture is placed with the methodology because it defines the sequence of player construction, scoring, attribution, intervention, and outcome measurement before the experimental results are introduced.

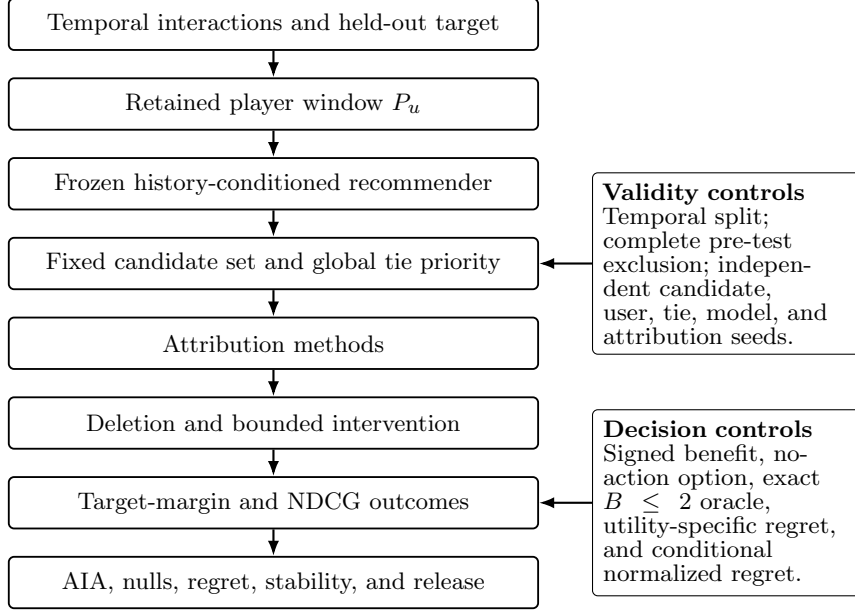


Fig. 1 ActionShap evaluation architecture. The fixed vertical path separates player construction, model scoring, attribution, intervention execution, and outcome measurement. Side panels show controls held constant across methods.

4.2 Attribution methods

For a sampled base permutation π_m , let S_{p,π_m} be the predecessor coalition of player p . We use M_{pair} base permutations and also evaluate each reverse order, giving $T = 2M_{\text{pair}}$ total prefix walks. The estimator is

$$\hat{\phi}_{u,p} = \frac{1}{T} \sum_{t=1}^T [v(S_{p,\pi_t} \cup \{p\}) - v(S_{p,\pi_t})]. \quad (13)$$

The exact value for reference is

$$\phi_{u,p}^{\text{exact}} = \sum_{S \subseteq P_u \setminus \{p\}} \frac{|S|! (n_u - |S| - 1)!}{n_u!} [v_u^{\text{attr}}(S \cup \{p\}) - v_u^{\text{attr}}(S)]. \quad (14)$$

The reverse-paired estimator above is unbiased for this factorial-weighted value; it is used because n_u can be too large for enumerating all 2^{n_u} coalitions. The reverse orders are reverse-paired samples used as a variance-reduction heuristic, and coalition states are cached. In the executable configuration, `permutations` denotes $M_{\text{pair}} = 500$, hence $T = 1000$ evaluated orders. The comparison includes locally weighted binary-mask LIME, LOO, greedy sequential deletion, and random attribution. No method observes measured intervention effects, oracle actions, or aggregate test outcomes.

Algorithm 1 Greedy sequential-deletion baseline

Require: target-margin utility v , player set P_u **Ensure:** signed ranking vector ϕ_u^{greedy}

- 1: $S \leftarrow P_u$; recompute $v(S)$
 - 2: **while** $S \neq \emptyset$ **do**
 - 3: evaluate every $v(S \setminus \{p\}) - v(S)$ for $p \in S$
 - 4: choose the maximum; break ties by lowest player index
 - 5: record its negative as $\phi_{u,p}^{\text{greedy}}$; set $S \leftarrow S \setminus \{p\}$
 - 6: **end while**
-

Algorithm 2 Random attribution negative control

Require: user u , player count n_u , seed s **Ensure:** random signed scores and optional action

- 1: draw $r_{u,p} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ for $p = 1, \dots, n_u$
 - 2: apply the common signed downweight rule with budget B and abstention
 - 3: do not evaluate intervention effects before selecting the action
-

For LIME, a binary mask $m_b \in \{0, 1\}^{n_u}$ defines coalition $S_b = \{p : m_{b,p} = 1\}$ and response $y_b = v_u^{\text{attr}}(S_b)$. With normalized Hamming distance $d_b = n_u^{-1} \sum_p (1 - m_{b,p})$, kernel width $\kappa = .25$, and ridge coefficient $\lambda_{\text{LIME}} = 1$, the fitted local surrogate is

$$(\hat{\beta}_0, \hat{\beta}) = \arg \min_{\beta_0, \beta} \sum_{b=1}^{512} e^{-d_b^2/\kappa^2} (y_b - \beta_0 - m_b^\top \beta)^2 + \lambda_{\text{LIME}} \|\beta\|_2^2. \quad (15)$$

The intercept is unpenalized and $\hat{\beta}$ is the LIME attribution. The mask design contains the full, empty, and every leave-one-out mask before seeded random masks fill the remaining rows.

For completeness, the greedy and random controls are defined operationally. Greedy sequential deletion starts at $S_0 = P_u$. At step t it evaluates $v_u^{\text{attr}}(S_{t-1} \setminus \{p\}) - v_u^{\text{attr}}(S_{t-1})$ for every remaining player, selects the largest benefit, breaks ties by the lowest player index, records the signed value under the convention that $-\phi$ predicts downweight benefit, and recomputes the next step from the reduced coalition. It repeats until every player is ordered; it does not inspect bounded effects or the oracle, and the later signed action rule may abstain. Random attribution draws independent scores $r_{u,p} \sim \mathcal{N}(0, 1)$ for every user and experiment seed using NumPy’s seeded generator. The random vector is passed through the same signed action-selection rule as every explainer, including the same budget and abstention option. Its seed is independently offset from the model/attribution seed by $1,000,000 + u$, so it is redrawn per user and seed.

4.3 Metrics

Let $\rho_S(\cdot, \cdot)$ denote Spearman rank correlation. Magnitude AIA is

$$\text{AIA}_u^z(g) = \rho_S(|\phi_{u,p}^g|, |\Delta_{u,p}^{z, \text{bnd}}|), \quad z \in \{\text{attr}, \text{NDCG}\}. \quad (16)$$

Deletion AIA substitutes $\Delta_{u,p}^{z,\text{del}}$; bounded AIA uses $\Delta_{u,p}^{z,\text{bnd}}$, and their difference is the Actionability Gap. Signed alignment is $\rho_S(-\phi_{u,p}^g, \Delta_{u,p}^z)$. For informative non-zero effects $I_u^z = \{p : \Delta_{u,p}^z \neq 0\}$, direction accuracy is

$$\text{DirAcc}_u^z(g) = |I_u^z|^{-1} \sum_{p \in I_u^z} \mathbf{1} [\text{sign}(-\phi_{u,p}^g) = \text{sign}(\Delta_{u,p}^z)]. \quad (17)$$

When $I_u^z = \emptyset$, direction accuracy is missing. Let $k_u = \min(k, n_u)$ and let T_{u,k_u}^g be the k_u players with largest predicted benefit $-\phi^g$; top- k intervention precision is

$$\text{Prec@}k_u^z(g) = k_u^{-1} \sum_{p \in T_{u,k_u}^g} \mathbf{1} [\Delta_{u,p}^z > 0 \wedge \text{sign}(-\phi_{u,p}^g) = \text{sign}(\Delta_{u,p}^z)]. \quad (18)$$

For a selected joint action $\hat{A}_{u,g}$, success is $\mathbf{1}[\Delta_u^z(\hat{A}_{u,g}) > 0]$; a negative realized effect identifies a harmful action, and abstention is $\mathbf{1}[\hat{A}_{u,g} = \emptyset]$. Rank stability over R_{seed} seed runs is

$$\text{Stab}_u(g) = \frac{2}{R_{\text{seed}}(R_{\text{seed}} - 1)} \sum_{r < s} \rho_S(\phi_{u,r}^{g,r}, \phi_{u,s}^{g,s}), \quad (19)$$

computed only over finite nonconstant pairs.

For each user and seed, $R_{\text{null}} = 1,000$ within-seed permutations shuffle effect magnitudes across players. We first compute the observed AIA and its R null draws within each seed; then we average the observed value and each matched null draw across the five seeds within user. Bootstrap and paired inference operate on these distinct-user summaries. We report null means, null 95th percentiles, plus-one permutation p -values, and valid-user counts. Constant vectors are missing, not imputed.

4.4 Decision quality and regret

For a method action, define the predicted joint benefit

$$\hat{B}_u^g(A) = - \sum_{p \in A} \hat{\phi}_{u,p}^g. \quad (20)$$

The method considers an action eligible when $\hat{B}_u^g(A) > 10^{-12}$; otherwise it abstains. Because this predicted benefit is additive, the implementation obtains the maximizer by sorting strictly positive individual benefits and taking at most B of them; equal scores use the deterministic tie rule below. For utility z , the exact budget-two oracle is

$$A_u^{*,z} = \arg \max_{A \in \mathcal{A}_{u,2}} \Delta_u^z(A), \quad (21)$$

and the method action is $\hat{A}_{u,g}$. Regret is

$$\text{Regret}_u^z(g) = \Delta_u^z(A_u^{*,z}) - \Delta_u^z(\hat{A}_{u,g}). \quad (22)$$

Algorithm 3 Paired Monte Carlo attribution

Require: $u, P_u, v, M_{\text{pair}}$, attribution seed

Ensure: $\hat{\phi}_u$

- 1: Initialise $\hat{\phi}_{u,p} \leftarrow 0$ for all $p \in P_u$
 - 2: **for** $m = 1$ to M_{pair} **do**
 - 3: Draw base order π_m and its reverse; evaluate both cached prefix walks
 - 4: **for** each consecutive pair $(S, S \cup \{p\})$ **do**
 - 5: $\hat{\phi}_{u,p} \leftarrow \hat{\phi}_{u,p} + v(S \cup \{p\}) - v(S)$
 - 6: **end for**
 - 7: **end for**
 - 8: Divide by $T = 2M_{\text{pair}}$ and record finite/constant-vector flags
-

For a general budget B , the utility-specific feasible oracle is

$$A_u^{*,z}(B) = \arg \max_{A \subseteq P_u, |A| \leq B} \Delta_u^z(A). \quad (23)$$

At $B = 2$ the released primary results exhaustively enumerate no action, all singletons, and all pairs. The archived $B = 3$ sensitivity was produced with the declared greedy lower-bound search rather than exhaustive triples; it is retained only for joint-effect, success, and abstention sensitivity. No normalized-regret claim is made for $B = 3$. Exhaustive triples would require at most $1 + 20 + \binom{20}{2} + \binom{20}{3} = 1351$ actions per user and are recommended for a future regenerated $B = 3$ release.

For utility z , normalized regret is

$$\text{NRegret}_u^z(g) = \frac{\Delta_u^z(A_u^{*,z}) - \Delta_u^z(\hat{A}_{u,g})}{\Delta_u^z(A_u^{*,z})}, \quad \Delta_u^z(A_u^{*,z}) > \varepsilon, \quad \varepsilon = 10^{-12}. \quad (24)$$

It is averaged only over active-oracle users. Zero means an oracle-equivalent action; one means that the oracle improves the utility but the selected action achieves no improvement; values above one mean that the selected action is harmful. Values below zero are rejected by validation for the exact shared action space (up to floating-point tolerance); they would indicate an oracle or action-space inconsistency. The quantity is not clipped to one.

Equal action scores use a deterministic common rule: maximize predicted benefit first; among ties choose the smaller action cardinality; among equal-cardinality ties choose the lexicographically lower sorted player-index tuple. An abstention action wins when the best predicted benefit is at most 10^{-12} . The exact oracle uses the same cardinality and lexicographic rule after maximizing realized utility.

5 Experimental Protocol

5.1 Datasets and temporal split

We use MovieLens-1M [17, 42] and Amazon Digital Music reconstructed from the Amazon Reviews 2018 release. Positive feedback is rating ≥ 4 ; Amazon duplicate

Algorithm 4 Signed action selection and exact budget-two oracle

Require: $\hat{\phi}_u$, profile weights, $\rho = 0.5$, candidate set

Ensure: Method action and utility-specific oracle actions

- 1: Enumerate $\emptyset$, every singleton, and every pair
 - 2: **for** each action A **do**
 - 3: Apply $w_p \leftarrow \rho w_p$ simultaneously for $p \in A$
 - 4: Cache target-margin and NDCG effects
 - 5: **end for**
 - 6: Compute $\hat{B}_u^g(A) = -\sum_{p \in A} \hat{\phi}_{u,p}$ for each candidate action
 - 7: Select the maximum eligible $\hat{B}_u^g(A)$ using the deterministic tie rule
 - 8: Return $\emptyset$ when the best score is at most 10^{-12}
 - 9: Select the exact oracle separately for each utility
-

Algorithm 5 Distinct-user inference and AIA null

Require: Five seed records per user and $R = 1000$ null draws per seed

Ensure: User-level estimates and corrected comparisons

- 1: **for** each distinct user u and seed s **do**
 - 2: Compute seed-level AIA only for finite, nonconstant vectors
 - 3: **for** $r = 1$ to R **do**
 - 4: Shuffle effect magnitudes within (u, s) ; recompute the null AIA
 - 5: **end for**
 - 6: **end for**
 - 7: **for** each distinct user u **do**
 - 8: Average the observed AIA and each matched null draw across seeds
 - 9: **end for**
 - 10: Bootstrap users; apply paired plus-one tests and Holm correction
-

user-item records retain the latest deterministic event and iterative 5-core filtering is reapplied after thresholding. MovieLens retains 6,035 users, 3,533 items, and 575,276 positive interactions. Amazon retains 10,623 users, 8,582 items, and 104,431 interactions. Source and processed hashes are recorded in the final manifest.

5.2 Models, candidates, and gates

The matrix includes ItemKNN and profile aggregation, five seeds, primary 1,000-user sampled ranking, 250-user full-catalogue subsets, and predeclared history, candidate, intervention-strength, budget, and utility sensitivities. ItemKNN uses cosine item-item similarities from binary training histories, removes self-similarity, retains the 200 largest non-zero neighbours per item, and applies no shrinkage or minimum-support threshold beyond the observed co-occurrence counts; all retained similarities are non-negative. The profile robustness model uses 64-dimensional item embeddings, ten leave-one-out BPR-style epochs, one sampled positive per user per epoch, uniform negatives outside the complete training history, learning rate 0.03, and L_2 coefficient 10^{-4} . Both models

Algorithm 6 Independent convergence selection

Require: $M_{\text{pair}} \in \{25, 50, 100, 250, 500, 1000\}$ and an independent $M_{\text{pair}} = 1000$ reference

Ensure: Selected M_{pair} or explicit unconverged status

- 1: **for** each candidate M_{pair} **do**
 - 2: For each candidate M_{pair} , evaluate $T = 2M_{\text{pair}}$ forward/reverse orders and compare attribution ranks and signed- $B = 2$ action Jaccard with the independent reference
 - 3: Record rank-valid fraction and threshold coverage
 - 4: **end for**
 - 5: Select smallest M_{pair} with rank $\geq .95$, Jaccard $\geq .80$, rank validity $\geq .95$
 - 6: **if** no candidate qualifies **then**
 - 7: Retain $M_{\text{pair}} = 1000$ as an unconverged stress test
 - 8: **end if**
-

recompute the profile from the retained player window at inference time and are not retrained under an intervention.

The primary candidate set contains the target and 199 uniformly sampled unseen negatives. Candidate seed 1729, user seed 2718, tie seed 31415, and model/attribution seeds 42–46 are independent. The gate uses 200 users and a fixed 200-item sampled candidate set. ItemKNN passes all blocking masking gates. The profile model fails the NDCG masking threshold for one MovieLens seed (mean absolute change 0.000682 versus 0.001) and is retained as a nonblocking robustness boundary.

LIME uses 512 binary masks including the full, empty, and all leave-one-out masks, a normalized Hamming-distance kernel with width 0.25, and ridge coefficient 1.0. The target-margin competitor count is $L = 10$ and the sigmoid temperature is one. The primary intervention is $\rho = 0.5$; $\rho = 0.25$ is sensitivity-only. The exact budget-two action space includes no action, all singletons, and all pairs.

5.3 Statistical analysis

For each seed we compute user-level quantities first; repeated seed values are then averaged within distinct user before bootstrap resampling. User bootstrap intervals are two-sided percentile intervals from 10,000 user resamples. For a paired metric, $d_u = x_u - y_u$ and $T_{\text{obs}} = n^{-1} \sum_{u=1}^n d_u$. Paired plus-one sign permutations use 10,000 two-sided sign flips $T_r^* = n^{-1} \sum_u \epsilon_{u,r} d_u$, with $\epsilon_{u,r} \in \{-1, +1\}$ and $p = (1 + \#\{|T_r^*| \geq |T_{\text{obs}}|\}) / (R + 1)$. Holm correction is applied separately within each dataset–model–condition–metric family across the declared method-pair comparisons. Paired Cohen’s d_z is $\bar{d}/s_d$, where d_u is the user-level paired difference and s_d its sample standard deviation. The paired comparison uses the intersection of users with finite values for both methods; missing correlations are retained as missing. Active-oracle counts define the denominator for normalized regret. No coalition evaluation is an independent statistical observation.

5.4 Reproducibility

The real result archive contains 80 recommendation runs plus five convergence studies. All assets in this manuscript are generated from schema-v2 raw results with repository-relative paths and content hashes. The primary ItemKNN target-margin analysis uses $M_{\text{pair}} = 500$ base permutations and $T = 1000$ evaluated orders; the independent convergence study selects lower diagnostic values for some conditions, while the conservative final floor is retained. The archived runs record macOS-26.5.1-arm64, Python 3.12.13, NumPy 2.4.6, SciPy 1.17.1, pandas 2.3.3, scikit-learn 1.9.0, matplotlib 3.11.1, and PyYAML 6.0.3. Historical processor model, RAM, and peak RSS were not serialized.

6 Results

All numerical claims below are generated only when `validation_report.json` reports PASS. The legacy pilot is excluded from the canonical package.

6.1 Recommendation quality and protocol audit

Table ?? shows that primary ItemKNN exceeds the popularity reference on both datasets. MovieLens ItemKNN reaches $\text{NDCG@10} = 0.284$ and $\text{HitRate@10} = 0.504$, compared with 0.158 and 0.320 for popularity. Amazon ItemKNN reaches 0.181 and 0.285, compared with 0.090 and 0.155. The profile model is weaker on both datasets and is therefore interpreted only as architectural robustness evidence.

Because each user has one held-out target, sampled-set HitRate@10 is numerically the same indicator as a one-target recall measure; HitRate is used here to avoid any implication of catalogue-level retrieval recall.

6.2 Faithfulness versus bounded alignment

Table 3 separates deletion AIA, bounded AIA, and their difference. On MovieLens ItemKNN, Shapley has bounded AIA 0.779 versus 0.933 for LIME and 0.978 for LOO. On Amazon, Shapley has bounded AIA 0.414 versus 0.827 and 0.851. Thus Shapley does not win absolute alignment.

The contrast is in the change between perturbations. Shapley’s primary gap is +0.0169 on MovieLens and +0.1285 on Amazon. LIME and LOO are negative. The random control is close to zero and its confidence intervals include zero in the primary conditions. Greedy is negative on the primary conditions. The gap therefore measures a change of evaluation target; it is not an explanation-validity score.

Primary 95% user-bootstrap CIs (deletion, bounded, gap): MovieLens Shapley [.751, .773], [.769, .789], [.013, .022]; LIME [.948, .954], [.928, .938], [−.022, −.014]; LOO [1.000, 1.000], [.974, .981], [−.026, −.019]; Random [−.008, .006], [−.006, .007], [−.001, .005]. Amazon Shapley [.248, .323], [.381, .448], [.110, .148]; LIME [.924, .937], [.811, .843], [−.119, −.089]; LOO [1.000, 1.000], [.835, .867], [−.166, −.133]; Random [−.023, .007], [−.025, .005], [−.010, .005].

6.3 Joint intervention decisions

The operational outcome is NDCG@10 under the exact $B = 2$ action space. On MovieLens ItemKNN, LOO yields mean NDCG effect 0.0431, LIME 0.0425, and Shapley 0.0403. Success

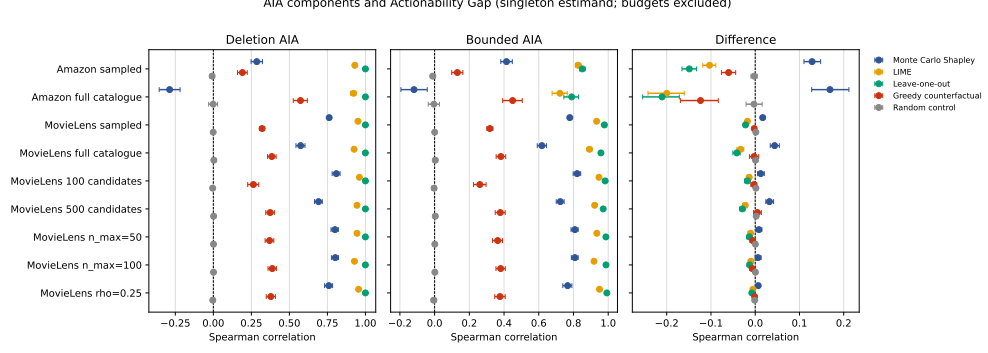


Fig. 2 Deletion AIA, bounded-intervention AIA, and their difference for the five methods. Points are distinct-user means and intervals are user-bootstrap 95% confidence intervals. Budget rows are excluded because budgets do not define singleton gap estimands.

Table 3 Primary ItemKNN alignment components. Values are distinct-user means. Primary 95% bootstrap intervals are reported in Section 6; complete intervals and missing-user counts are provided in the release.

Dataset	Method	Deletion AIA	Bounded AIA	Gap	Valid n
MovieLens	Monte Carlo Shapley	0.762	0.779	0.017	1000
MovieLens	LIME	0.951	0.933	-0.018	1000
MovieLens	LOO	1.000	0.978	-0.022	1000
MovieLens	Greedy	0.321	0.318	-0.002	1000
MovieLens	Random	-0.001	0.000	0.001	1000
Amazon	Monte Carlo Shapley	0.286	0.414	0.129	993
Amazon	LIME	0.930	0.827	-0.103	993
Amazon	LOO	1.000	0.851	-0.149	993
Amazon	Greedy	0.191	0.131	-0.060	993
Amazon	Random	-0.008	-0.010	-0.002	993

is 28.9%, 28.8%, and 27.4%, respectively. Conditional normalized regret among 339 active-oracle users is 0.217, 0.225, and 0.268. Local methods therefore retain a modest decision advantage on MovieLens.

On Amazon ItemKNN, NDCG effects are 0.0333 for Shapley, 0.0331 for LIME, and 0.0313 for LOO. Conditional normalized regret among 196 active-oracle users is 0.206, 0.197, and 0.232. The differences are small and do not support a universal winner. Random actions have negative mean NDCG effects on both datasets.

Primary NDCG-effect 95% user-bootstrap CIs: MovieLens Shapley [.035, .046], LIME [.037, .049], LOO [.037, .050]; Amazon Shapley [.027, .040], LIME [.027, .039], LOO [.026, .037]. NRegret CIs and all valid/missing counts remain in the machine-readable release.

6.4 Null calibration and convergence

The null-calibration table below reports the complete within-user null alongside the component table. The within-user AIA null is centred near zero for all primary ItemKNN conditions. Shapley, LIME, LOO, and greedy exceed the null for absolute AIA, while random attribution

Table 4 Primary budget-two intervention decisions. NRegret is conditional on active NDCG oracles; the active-user count is shown in parentheses.

Dataset	Method	ΔNDCG	Success	Abstention	NRegret (n)
MovieLens	Shapley	0.040	27.4%	0.0%	0.268 (339)
MovieLens	LIME	0.043	28.8%	0.0%	0.225 (339)
MovieLens	LOO	0.043	28.9%	0.0%	0.217 (339)
MovieLens	Greedy	0.006	10.3%	0.0%	1.471 (339)
MovieLens	Random	-0.001	7.4%	0.0%	1.446 (339)
Amazon	Shapley	0.033	17.0%	1.0%	0.206 (196)
Amazon	LIME	0.033	17.1%	7.0%	0.197 (196)
Amazon	LOO	0.031	16.7%	8.6%	0.232 (196)
Amazon	Greedy	0.010	9.0%	0.7%	0.784 (196)
Amazon	Random	-0.004	6.6%	4.8%	1.197 (196)

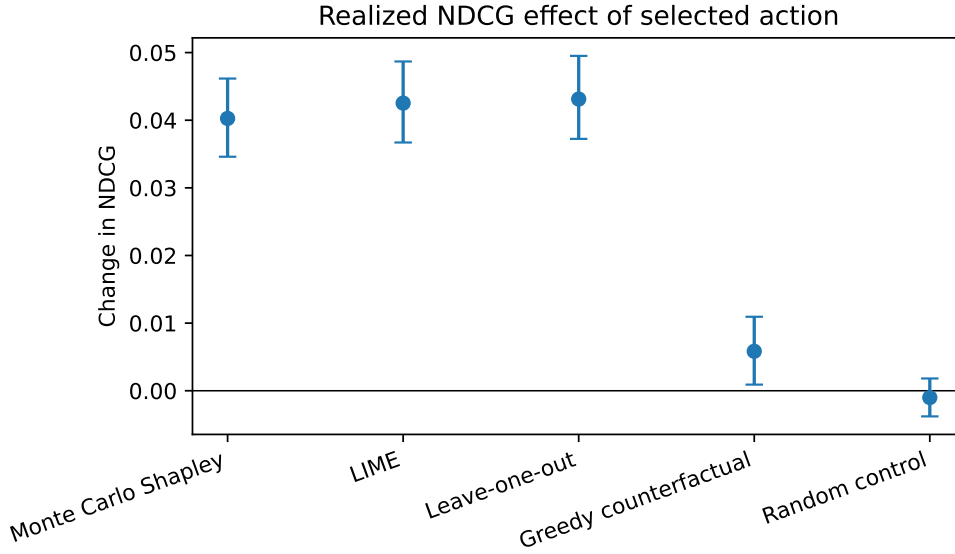


Fig. 3 Primary MovieLens ItemKNN NDCG effect for actions selected at budget two. We show MovieLens for visual legibility; the analogous Amazon effects are reported in Table 4. The comparison is retrospective and target-conditioned; intervals are distinct-user bootstrap intervals.

does not. This establishes that absolute alignment is not merely a consequence of the correlation statistic returning positive values.

Target-margin convergence selects $M_{\text{pair}} = 50$ for Amazon ItemKNN, $M_{\text{pair}} = 100$ for Amazon profile, $M_{\text{pair}} = 250$ for MovieLens ItemKNN, and $M_{\text{pair}} = 50$ for MovieLens profile; the final suite retains the conservative floor $M_{\text{pair}} = 500$ ($T = 1000$ evaluated orders). Rank-valid fraction is the fraction of users with finite rank comparisons, whereas threshold coverage is the fraction of rank-valid users satisfying both the rank and action-overlap thresholds; coverage is diagnostic and is not a replacement for the rank-validity criterion. The NDCG

Table 6 Selected target-margin Monte Carlo convergence budgets. Only selected rows are shown. M_{pair} counts base permutations and T counts evaluated forward/reverse orders.

Dataset	Model	M_{pair}	T	Rank	Jaccard	Rank-valid	Coverage
Amazon	itemknn	50	100	0.9532	0.9090	100.0%	60.0%
Amazon	profile	100	200	0.9670	0.9110	100.0%	66.0%
MovieLens	itemknn	250	500	0.9853	0.8433	100.0%	69.0%
MovieLens	profile	50	100	0.9670	0.8320	100.0%	56.0%

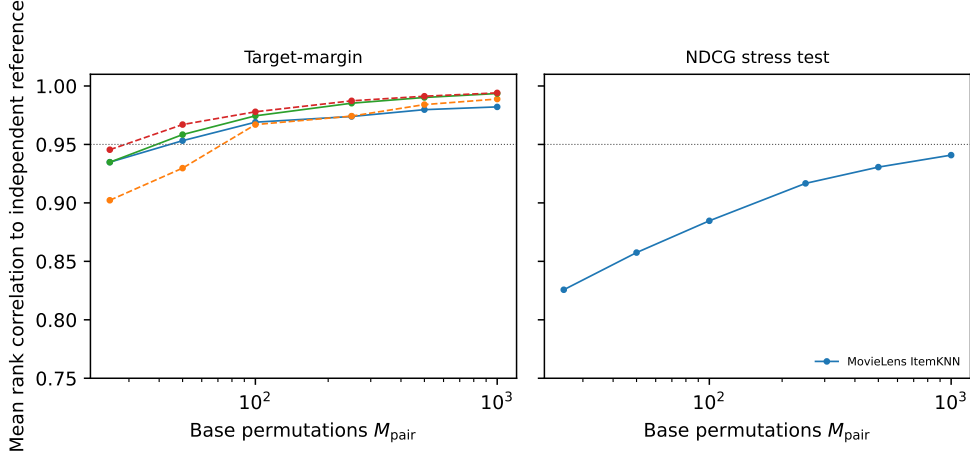


Fig. 4 Independent Monte Carlo convergence against an $M_{\text{pair}} = 1000$ ($T = 2000$ orders) reference. Efficiency is not used as a convergence criterion because the prefix walk telescopes.

attribution sensitivity remains unconverged at $M_{\text{pair}} = 1000$ and is reported only as a stress test.

6.5 Robustness and boundaries

The positive Shapley gap persists across the primary ItemKNN candidate, history, intervention-strength, and full-catalogue conditions, but the absolute values vary. For Amazon full catalogue, the positive difference reflects reduced anti-alignment rather than positive bounded-intervention alignment: deletion AIA is -0.2875 and bounded AIA is -0.1186 , so both absolute AIA values remain negative. Full-catalogue NDCG alignment is sparse and is treated descriptively. The longer-history conditions also show reduced masking sensitivity, supporting the use of $n_{\text{max}} = 20$ as a declared operating boundary rather than an arbitrary computational cap.

The profile model supplies a useful negative boundary: it can change rankings while underperforming popularity, and one MovieLens seed fails the NDCG masking threshold. This demonstrates why a non-degenerate masking gate and recommendation-quality audit are necessary before interpreting attribution results.

6.6 Computational cost and release provenance

Table 7 reports the allocation-stage computational budgets. These are analytical scorer-call counts, not claims that every call has identical wall-clock cost; the exact ItemKNN batch

Table 7 Attribution-stage computational cost for a user with $n_u \leq 20$ players. Joint-action selection and the common exact oracle are excluded because they are shared outcome evaluations.

Method	Allocation-stage score budget	Reproducible definition
Monte Carlo Shapley	$T(n_u + 1) \leq 21,000$ prefix states before cache reuse	$M_{\text{pair}} = 500$ base orders plus reverses; coalition states cached and batched.
LIME	512 binary masks	Full, empty, all leave-one-out masks, then seeded random masks; weighted ridge with Hamming kernel.
LOO	$n_u + 1 \leq 21$ states	Full coalition and every one-deletion coalition; exact deletion diagnostic.
Greedy sequential deletion	$1 + n_u(n_u + 1)/2 \leq$ 211 states	Every remaining deletion is recomputed at every step; lower-index ties win.
Random control	0 attribution utility calls	IID $\mathcal{N}(0, 1)$ scores; the common action/effect evaluation is still applied after selection.

implementation reuses cached similarity blocks, while the joint-action oracle is evaluated once per user and shared across methods.

The canonical artifact directory is `paper-ideas/ActionShap/paper-v3/`. The reproducibility command is:

```
python scripts/make_paper_assets.py --raw results/raw --out ../paper-v3
```

The schema-v2 archive records total wall-clock duration for the 80 recommendation runs: the 60 ItemKNN runs range from approximately 29.8 to 2,851.6 seconds (mean 183.0 seconds), and the 20 profile runs range from approximately 26.9 to 299.6 seconds (mean 95.0 seconds). These are complete-pipeline durations rather than isolated explainer timings; the analytical call-budget table above is the comparable per-method cost measure. The archived runs record macOS-26.5.1-arm64, Python 3.12.13, NumPy 2.4.6, SciPy 1.17.1, pandas 2.3.3, scikit-learn 1.9.0, matplotlib 3.11.1, and PyYAML 6.0.3. Processor model, RAM, and peak RSS were not serialized in the historical schema-v2 files, so no unsupported hardware claim is inferred; the current runner records those fields for any regenerated release. The schema-v2 result archive SHA-256 is `ac4c7fb1993458b6b41054974ebff215710e7a8b5894c7aa6af828e94b2a5b0f`.

All paths in generated manifests are repository-relative and all release archives are verified by SHA-256. Code, configuration files, validation reports, preprocessing/run scripts, and machine-readable result matrices are included in the package’s `release/` directory and the repository release. This manuscript and supplement are generated from source commit `b778f55`; the schema-v2 result archive has SHA-256 `ac4c7fb1993458b6b41054974ebff215710e7a8b5894c7aa6af828e94b2a5b0f`. The source repository URL is <https://github.com/mouadlouhichi/next-paper>; its public visibility and any DOI-backed deposit remain external release actions.

7 Discussion and Broader Implications

The results support a three-axis interpretation. LIME and LOO are strongest on absolute pointwise alignment because their constructions are local to deletion or the full profile. Shapley has lower absolute alignment but a larger positive change when the evaluation moves from deletion to bounded downweighting. This is consistent with coalition-context averaging, but the experiment does not establish a causal mechanism and the random control prevents interpreting the gap as standalone validity.

Table 8 compares the protocol dimensions most relevant to the closest counterfactual and explanation-evaluation studies. ActionShap differs not by claiming a new causal estimator, but

Table 8 Component-level positioning against the closest related evaluation and counterfactual recommendation work.

Work	Players	Intervention/policy	Outcome	Joint oracle	Inference
PRINCE [22]	Provider/profile	Counterfactual generation	Recommendation change	Method-specific	Method-specific
ACCENT [23]	User factors	Counterfactual masking	Top- K explanation	No common budget oracle	Method-specific
Tan et al. [24]	Recommendation factors	Counterfactual explanation	Ranking change	Method-specific	User/item evaluation
CEERS [26]	Explanation factors	Counterfactual evaluation	Explanation quality	Benchmark-specific	Reproducibility protocol
Barkan et al. [27]	Learned masks	Model-specific counterfactual	Explanation quality	Learned objective	Method-specific
LXR [28]	Learned explanation masks	Model-specific counterfactual	Explanation quality	Learned objective	Method-specific
Mohammadi et al. [29]	Explanation factors	Top- K counterfactual evaluation	Consistency	No common budget oracle	Benchmark study
Nguyen et al. [30]	User/profile factors	Top- K counterfactual evaluation	Effectiveness, sparsity	Benchmark-specific	Cross-paper benchmark
ActionShap	History interactions	$w_p \leftarrow \rho w_p$	AIA, NDCG, regret	Exact $B = 2$; abstention	Distinct-user paired inference

by combining interaction-player attribution, a declared bounded profile policy, a utility-specific joint oracle, abstention, and distinct-user inference.

The decision results reinforce this distinction. On MovieLens, local methods select slightly better NDCG actions. On Amazon, Shapley and LIME are close, with modest differences in conditional regret. A method can therefore show a positive change between deletion and bounded-intervention alignment without selecting the best action for every operational utility. This is precisely why ActionShap reports pointwise alignment, signed direction, success, abstention, and regret together.

The protocol also clarifies what offline actionability means. The intervention is system-executable inside the frozen simulator: it changes an inference-time interaction weight without retraining. It is not automatically a user action, causal intervention, or prospective policy. A production deployment would need authority, cost, user-consent, and feedback-loop specifications that are outside this benchmark.

7.1 Limitations and Threats to Validity

7.1.1 Construct validity

Actionability is defined as predictive validity under one bounded downweighting policy. It is not perceived usefulness, legal recourse, user agency, or a causal effect in the world. All primary players are technically mutable, so feasibility does not differentiate factors; future work should study heterogeneous costs and user-side constraints.

7.1.2 Metric validity

NDCG is discrete and produces constant effect vectors for some users. We report missing correlations rather than imputing zero. Target margin is the converged attribution utility but has less direct operational meaning; NDCG is the action outcome and is never relabelled as target margin.

7.1.3 Internal validity

Temporal splitting and complete pre-test exclusion reduce leakage, but the audit remains target-conditioned. The held-out target defines the labelled instance being explained, which is appropriate for retrospective evaluation but unavailable to a prospective policy. Candidate sets are fixed and target coverage is one by construction; sampled ranking is not retrieval evaluation.

7.1.4 External validity

Two datasets and two history-conditioned models do not cover sequential transformers, graph recommenders, multimodal inputs, online feedback, provider objectives, or heterogeneous intervention costs. The full-catalogue subset is smaller than the primary cohort. The profile model’s quality shortfall and one failed gate are retained as explicit boundaries rather than generalized.

8 Conclusion

ActionShap provides a reproducible protocol for asking whether recommendation explanations predict the effects of declared system-executable profile interventions. In the primary ItemKNN experiments, Shapley exhibits a positive bounded-minus-deletion alignment difference, particularly on Amazon, while LIME and LOO retain higher absolute alignment and slightly better MovieLens NDCG decisions. The random control remains near its chance null, and a positive Actionability Gap is therefore not sufficient evidence of explanation validity. The principal contribution is the separation of deletion faithfulness, bounded-intervention alignment, and joint decision quality under a common, leakage-safe, abstention-aware protocol.

Declarations

Funding. The authors declare that no funds, grants, or other support were received during the preparation of this manuscript.

Competing interests. The authors declare no competing interests.

Ethics approval. Not applicable. The study uses publicly available secondary recommendation data and involves no recruited human participants. Any future online or user study would require independent ethics review.

Consent to participate / for publication. Not applicable.

Data availability. MovieLens-1M was obtained from the GroupLens public release, and Amazon Digital Music was reconstructed from the Amazon Reviews 2018 public release after the documented rating threshold, deterministic deduplication, and iterative 5-core filtering. Source and processed-data SHA-256 values, split rules, and candidate construction are recorded in the final provenance manifest. The datasets are subject to their respective maintainers’ terms and are not redistributed in this repository.

Code and materials availability. The canonical submission package is `paper-ideas/ActionShap/paper-v3/`. The raw experiment archive is distributed separately through the content-addressed release workflow because source datasets and raw JSON files are not ordinary manuscript assets.

Author contributions. M.L.: conceptualization, methodology, software, formal analysis, investigation, and writing—original draft. R.N.: methodology, validation, and writing—review and editing. M.Laz.: supervision, methodology, and writing—review and editing. All authors approved the final manuscript.

A Formal Guarantees

A.1 Efficiency of the prefix-walk estimator

For any permutation $\pi = (p_1, \dots, p_n)$, define $S_0 = \emptyset$ and $S_j = \{p_1, \dots, p_j\}$. The prefix-walk contribution of p_j is $v(S_j) - v(S_{j-1})$. Consequently,

$$\sum_{j=1}^n [v(S_j) - v(S_{j-1})] = v(P_u) - v(\emptyset). \quad (\text{A1})$$

Averaging over paired permutations preserves the identity up to floating-point arithmetic. The efficiency residual in the release is therefore a cache and arithmetic diagnostic, not a convergence criterion.

A.2 Leave-one-out ceiling

For deletion at $\rho = 0$, $\Delta_{u,p}^{\text{del}} = v(P_u \setminus \{p\}) - v(P_u)$ and $\phi_{u,p}^{\text{LOO}} = v(P_u) - v(P_u \setminus \{p\})$. Thus $|\phi_{u,p}^{\text{LOO}}| = |\Delta_{u,p}^{\text{del}}|$ for every valid nonconstant user and deletion AIA is one. LOO's deletion-to-bounded gap is consequently bounded above by zero; it is retained as a diagnostic oracle, not a positive-gap competitor.

A.3 Exact budget-two action space

With $n_u \leq 20$, the exact action space contains $1 + n_u + \binom{n_u}{2} \leq 211$ actions, including no action. Each action is evaluated against target margin and NDCG using the same cached candidate scores. The oracle is therefore exact for every primary user and does not rely on a sampled action subset.

B Experimental Contract

The profile optimizer samples one positive item per user per epoch, excludes that positive from its temporary context, and samples a negative uniformly outside the user’s complete training history. ItemKNN has no latent user embedding, shrinkage coefficient, minimum-support cutoff, or negative-similarity branch: the retained similarity matrix is non-negative by construction. AIA is Spearman rank correlation and is missing for constant or non-finite vectors. Normalized regret uses $\varepsilon = 10^{-12}$ and is averaged only over positive-oracle users. The release archive records the exact dependency versions, source hashes, run durations, and selected users; complete per-user vectors remain in the compressed machine-readable asset rather than in the PDF.

Table C1 Alignment robustness across the predeclared ItemKNN conditions. Deletion and bounded columns are Shapley; LIME and LOO columns are their Actionability Gaps.

Condition	Shapley Del.	Shapley Bnd.	Shapley Gap	LIME Gap	LOO Gap	Valid n
Amazon sampled	.286	.414	+.129	-.103	-.149	993
Amazon full catalogue	-.288	-.119	+.169	-.200	-.211	250
MovieLens sampled	.762	.779	+.017	-.018	-.022	1000
MovieLens full catalogue	.574	.618	+.044	-.033	-.042	250
MovieLens 100 candi- dates	.809	.821	+.012	-.014	-.018	250
MovieLens 500 candi- dates	.692	.724	+.032	-.023	-.029	250
MovieLens $n_{\max} = 50$	.800	.808	+.008	-.010	-.013	250
MovieLens $n_{\max} = 100$	.802	.808	+.006	-.010	-.013	250
MovieLens $\rho = .25$	.760	.766	+.007	-.005	-.008	250

Table C2 Decision robustness on primary and representative boundary conditions. NRegret is conditional on active NDCG oracles; B=3 has no normalized-regret claim.

Condition	Method	ΔNDCG	Success	Abstention	NRegret	Active n
MovieLens pri- mary	Shapley	.040	27.4%	0.0%	.268	339
MovieLens pri- mary	LIME	.043	28.8%	0.0%	.225	339
MovieLens pri- mary	LOO	.043	28.9%	0.0%	.217	339
Amazon primary	Shapley	.033	17.0%	1.0%	.206	196
Amazon primary	LIME	.033	17.1%	7.0%	.197	196
Amazon primary	LOO	.031	16.7%	8.6%	.232	196
MovieLens full catalogue	Shapley	.012	6.7%	—	.291	20
Amazon full cata- logue	Shapley	.003	1.5%	—	.256	5
$B = 1$	Shapley	.026	—	—	—	—
$B = 1$	LIME	.026	—	—	—	—
$B = 1$	LOO	.027	—	—	—	—
$B = 3$	Shapley	.062	—	—	—	—
$B = 3$	LIME	.062	—	—	—	—
$B = 3$	LOO	.061	—	—	—	—

C Robustness Results

Across these conditions, the Random gap remains approximately between $-.003$ and $+.002$, while Greedy ranges from approximately $-.124$ to $+.005$. The archived B=3 sensitivity uses greedy rather than exhaustive triple enumeration and therefore reports no normalized regret.

Table D1 Confirmatory paired comparisons. Bootstrap intervals are descriptive and unadjusted; significance is determined by Holm-adjusted permutation p -values.

Dataset	Metric	Comparison	Difference	Unadjusted CI	Holm p	d_z
Amazon	Gap	Shapley LIME	— .232	[.202,.263]	.0010	.473
Amazon	Gap	Shapley — LOO	.277	[.245,.311]	.0010	.525
Amazon	Bounded AIA	Shapley LIME	— -.413	[-.449,-.378]	.0010	-.726
Amazon	Δ NDCG	Shapley LIME	— .000	[-.002,.002]	.8983	.004
MovieLens	Gap	Shapley LIME	— .034	[.028,.043]	.0010	.289
MovieLens	Gap	Shapley — LOO	.039	[.032,.047]	.0010	.319
MovieLens	Bounded AIA	Shapley LIME	— -.155	[-.164,-.145]	.0010	-1.010
MovieLens	Δ NDCG	Shapley LIME	— -.002	[-.004,-.001]	.0144	-.085

Table D2 Selected convergence budgets and the unconverged NDCG stress test. Coverage means threshold coverage among rank-valid users.

Dataset	Model	Utility	M_{pair}	Rank	Jaccard	Rank-valid	Coverage
Amazon	ItemKNN	Margin	50	.953	.909	100%	60%
Amazon	Profile	Margin	100	.967	.911	100%	66%
MovieLens	ItemKNN	Margin	250	.985	.843	100%	69%
MovieLens	Profile	Margin	50	.967	.832	100%	56%
MovieLens	ItemKNN	NDCG	1000	.941	.781	92%	62%

D Statistical Validation and Convergence

Full pairwise comparisons remain in the content-addressed machine-readable release. The NDCG row is a declared stress test and is not used to select the primary target-margin budget.

Harm is not promoted as a separate aggregate headline metric; harmful selected actions are represented by negative realized effect and success is defined by a positive realized effect.

Table D3 Advertised diagnostic metrics for the primary ItemKNN target-margin analysis. Direction accuracy and top-3 precision use signed bounded effects; stability is rank stability across the five seed runs.

Dataset	Method	Direction accuracy	Precision@3	Stability	Valid n
MovieLens	Shapley	.888	.820	.985	1000
MovieLens	LIME	.963	.920	.975	1000
MovieLens	LOO	.989	.967	1.000	1000
MovieLens	Greedy sequential deletion	.617	.296	1.000	1000
MovieLens	Random	.502	.165	-.001	1000
Amazon	Shapley	.784	.603	.971	993
Amazon	LIME	.936	.596	.969	993
Amazon	LOO	.941	.596	1.000	993
Amazon	Greedy sequential deletion	.543	.361	1.000	993
Amazon	Random	.507	.277	-.010	993

Table E1 Computational cost and release record. Analytical budgets are per user; runtime values are complete-pipeline measurements.

Item	Value
Shapley budget	$T(n_u + 1) \leq 21,000$ prefix states before cache reuse.
LIME budget	512 masks.
LOO budget	$n_u + 1 \leq 21$ states.
Greedy budget	$1 + n_u(n_u + 1)/2 \leq 211$ states.
ItemKNN runtime	29.8–2851.6 s; mean 183.0 s over 60 runs.
Profile runtime	26.9–299.6 s; mean 95.0 s over 20 runs.
Environment	macOS-26.5.1-arm64; Python 3.12.13; NumPy 2.4.6; SciPy 1.17.1; pandas 2.3.3; scikit-learn 1.9.0; matplotlib 3.11.1; PyYAML 6.0.3.
Release	80 recommendation runs; 5 convergence studies; result archive SHA-256 is recorded in the release metadata.

E Computational Cost and Reproducibility

Complete per-user, per-seed, robustness, convergence, and pairwise-comparison matrices are provided in the content-addressed CSV/JSON release. The historical schema did not record processor model, RAM, or peak RSS; the current runner records those fields for future regenerated releases.

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